



Individual Project Report and Future Work Proposal

Automated Casting Defect Detection Using Deep Learning and Computer Vision

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Module	Advanced Artificial Intelligence Projects in Data Science	Code	55-710603
Leader	Royce Copley	Submission	18/08/2026

Part A: Introduction

This individual report presents the group project Automated Casting Defect Detection Using Deep Learning and Computer Vision for the module Advanced Artificial Intelligence Projects in Data Science. The project addresses a manufacturing quality-control problem: casting parts can contain visible surface defects, but manual inspection is slow, repetitive and affected by human fatigue. The system classifies top-view submersible pump impeller images as `def_front` or `ok_front` and demonstrates how AI can support quality inspectors with faster image-based screening.

My role in the team was evaluation, grad-cam & ethics lead, with a focus on final metrics, confusion matrix, error analysis, explainability and responsible AI. This individual report therefore explains the whole project while also highlighting my specific contribution to the final AI project package.

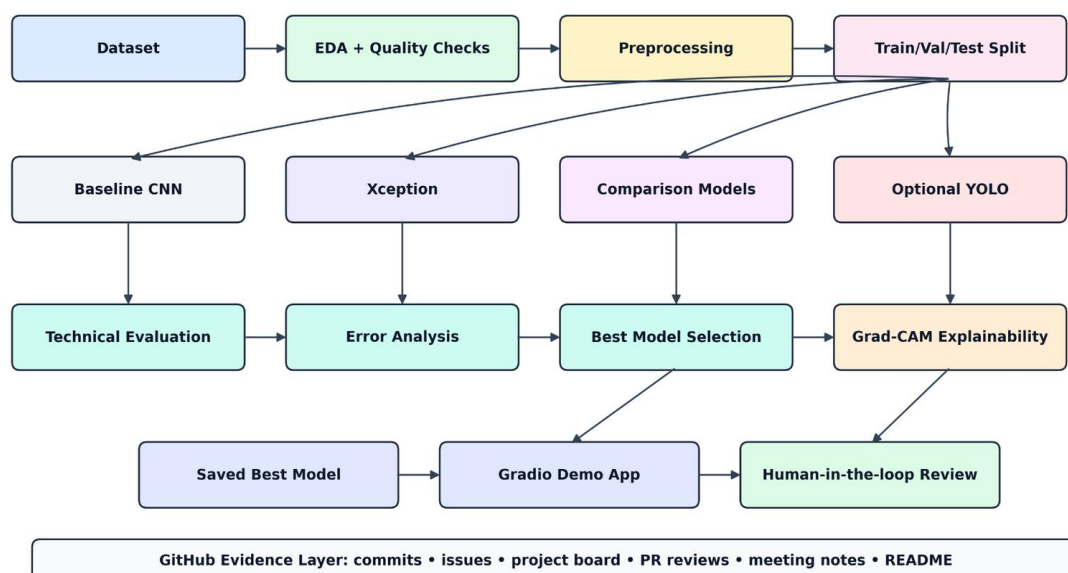
Part B: Technical Review (LO1)

The technical review focused on modern computer-vision methods for image classification. A Custom CNN was used as a baseline because it learns features directly from the project data and provides a simple comparison point. Transfer-learning models were then explored, including MobileNetV2, EfficientNetB0, DenseNet121 and Xception. Transfer learning was suitable because pretrained convolutional backbones already contain useful visual feature detectors, while the project dataset is smaller than the datasets normally required to train a deep CNN fully from scratch. Xception became the main model direction because its depthwise-separable convolution design is efficient and its validation evidence was strong. Grad-CAM was also considered because a quality-inspection system should not only output a class label, but should provide some visual support for why a decision was made.

Part C: Design and Implementation (LO2)

The dataset was obtained from Kaggle and contains casting manufacturing product images. The full end-to-end notebook used 7,348 augmented grayscale images organised into train and test folders, with `def_front` and `ok_front` subfolders. The training folder contained 3,758 defective and 2,875 acceptable images, while the test folder contained 453 defective and 262 acceptable images. A further unaugmented 512 x 512 version contains 781 `def_front` and 519 `ok_front` images. The workflow included image manifest creation, EDA, corrupted-image checking, duplicate checking, preprocessing, train-validation-test usage, class weighting, model training, model comparison, final testing, error analysis, model saving and demo preparation.

Casting Defect Detection - Clean System Architecture



My individual contribution focused on final evaluation, explainability and responsible AI. I reviewed the final classification results, interpreted the confusion matrix, considered the impact of the single misclassified image, and supported the ethical discussion around false negatives, human oversight and dataset limitations. This contribution was essential because the assessment requires both technical performance evaluation and social/ethical evaluation.

Part D: Technical and Critical Evaluation (LO3/LO4)

The strongest result in the full end-to-end notebook was the fine-tuned Xception model. On the held-out test set, Xception_fine_tuned achieved 99.86% accuracy, 0.9962 precision, 1.0000 recall, 0.9981 F1-score and 1.0000 AUC, with only one misclassified image out of 715 test images. These results show that the model learned strong visual differences between defective and acceptable casting images in the provided dataset. However, the EDA also found 128 duplicate file entries across 64 duplicate groups, so the high performance should be interpreted carefully. The responsible conclusion is that the system is a strong academic prototype, but it requires external validation on unseen factory data before real deployment.

Metric	Final Xception_fine_tuned result
Accuracy	99.86%
Precision	0.9962
Recall	1.0000
F1-score	0.9981
AUC	1.0000
Misclassified	1 out of 715

For my role, the confusion matrix is the most important evidence. It shows 452 correctly identified defective images and 262 correctly identified OK images, with only one defective image predicted as OK. This confirms strong performance but also highlights why false negatives must be managed responsibly. The ethical recommendation is to use the system with confidence thresholds and human review, especially for borderline cases.

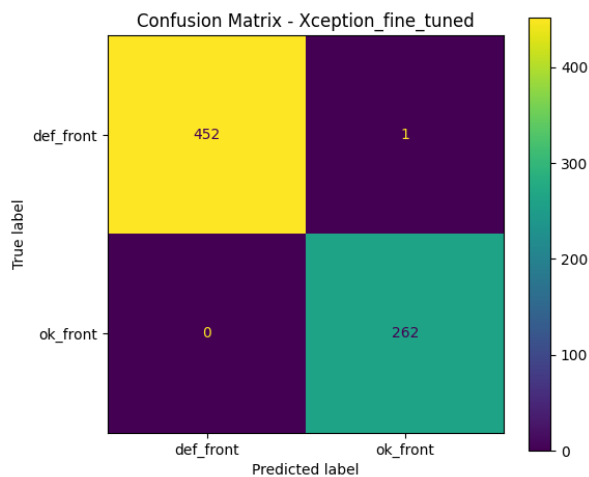


Figure 2. Final confusion matrix for Xception_fine_tuned on the test set.



Figure 3. Error-analysis screenshot showing the single misclassified test image.

The main social and ethical issue is the risk of over-reliance. A false negative could allow a defective product to pass inspection, which may affect cost, quality and customer trust. For that reason, the proposed role of the system is decision support rather than full automation. Low-confidence cases, unusual images and suspicious Grad-CAM patterns should be reviewed by a human inspector. Since no human participants or personal data were collected, the data-ethics risk is lower than in personal-data projects, but responsible research still requires transparency, limitation reporting, monitoring and human oversight.

Part E: Project Management and Future Work Proposal

The project was managed as a 12-week group development cycle. Weekly milestones, GitHub issues, commits, member contribution evidence, meeting notes and project-management trackers were used to show continuous progress. The strongest project-management practice was role clarity: each member had ownership of a defined project area while still contributing to shared decisions.

Week	Contribution evidence
Week 3	Evaluated baseline metrics.
Week 5	Prepared interpretation of model-comparison results.
Week 6	Prepared final test metrics and confusion matrix.
Week 7	Completed error analysis and responsible AI review.

The main challenges were duplicate-image awareness, choosing metrics beyond accuracy, fixing explainability for the nested Xception model, and preparing a reliable live demo environment. These were handled through critical limitation reporting, model comparison, issue tracking and backup planning.

Future work should collect new images from different production lines, lighting conditions, camera positions and defect types. The project could also be extended from binary classification to localisation or segmentation, so that the system identifies exact defect regions rather than only predicting a class. A practical deployment would need a stable camera setup, a monitored Flask or Gradio interface, audit logs, confidence thresholds, periodic model review and clear human-in-the-loop escalation.

Part F: Conclusion

The project delivered a complete AI quality-inspection pipeline: data understanding, preprocessing, baseline modelling, transfer learning, model comparison, final testing, explainability, demo preparation and GitHub evidence. My individual contribution supported the project area assigned to my role and helped connect the technical system to the assessment learning outcomes. The final model result is strong on the provided dataset, but responsible deployment requires external validation and human-in-the-loop quality review.

GitHub Source code: <https://github.com/likith9059/AdvancedArtificial-IntelligenceProjectsInDataScience-55-710603->

AI Declaration

ChatGPT was used as a supporting tool at selected stages of this project. Its use primarily involved gaining a clearer understanding of the dataset requirements, obtaining guidance on how graphs and charts could be understood, and exploring possible ideas during the planning of the system architecture. The suggestions provided by ChatGPT were used for guidance and learning purposes and were independently considered before being applied. All project decisions, development work, analysis, and final outcomes were completed and evaluated by the author.

References

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